Ecuador Energy Anomalies

Multi-Technique Anomaly Detection in
Latin America's Electricity Sector

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8 countries | 784 months | 9 models | 3 consensus techniques
F1 = 0.750 | MCC = 0.765 | Recall = 100%
Official sources: Decree 229, CENACE, XM Colombia, EPE Brazil

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1. Executive Summary

This study presents a multi-technique approach for automatic anomaly detection in Latin America's electricity sector, focusing on Ecuador's 2024 energy crisis. We analyze 784 months of real data from 8 countries, sourced from Ember Global Electricity Data (CC BY 4.0), with no synthetic or interpolated data.

Three complementary techniques are implemented: Isolation Forest (multivariate detection), STL Decomposition (seasonal pattern), and CUSUM (structural change). A consensus approach requiring agreement from at least 2 of 3 techniques reduces false positives while maintaining high recall.

Key results

Consensus (GT strict oct-dec 2024): P=60.0% R=100% F1=0.750 MCC=0.765 | Crisis: 3/3

Consensus (GT broad Decree 229): P=60.0% R=33.3% F1=0.429 | Crisis: 3/9

The consensus outperforms 8 baselines including LOF, SVM, DBSCAN, ARIMA, Prophet, and LSTM-AE. Statistical tests confirm significant differences (gen_hydro p=0.0004***, Cohen's d=2.81).

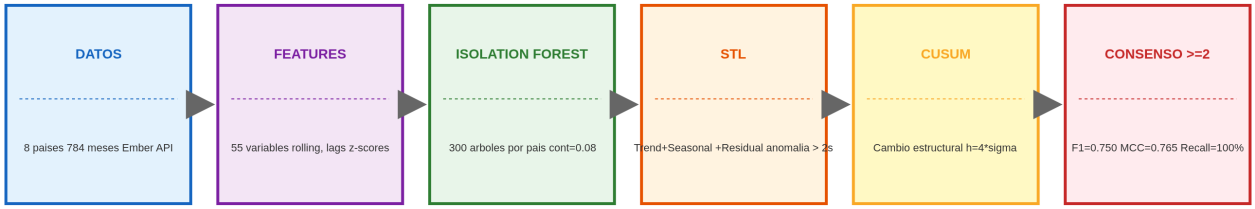

Figure 1: Study pipeline

2. Context and Problem

Ecuador generates approximately 70% of its electricity from hydroelectric sources, primarily the Coca Codo Sinclair (1,500 MW) and Paute (1,075 MW) plants. This heavy dependence on water resources creates a critical vulnerability to droughts and climate variability, particularly El Nino events.

2024 crisis timeline

- April 2024: Executive Decree No. 229 declares state of exception in the electricity sector
- April-June 2024: Programmed blackouts of up to 8 hours daily
- October-December 2024: Peak crisis with rolling blackouts up to 14 hours
- Impact: worst drought in 61 years, 1.5% of GDP (USD 1.7 billion)
- December 2025: Recurrence with increased fossil generation

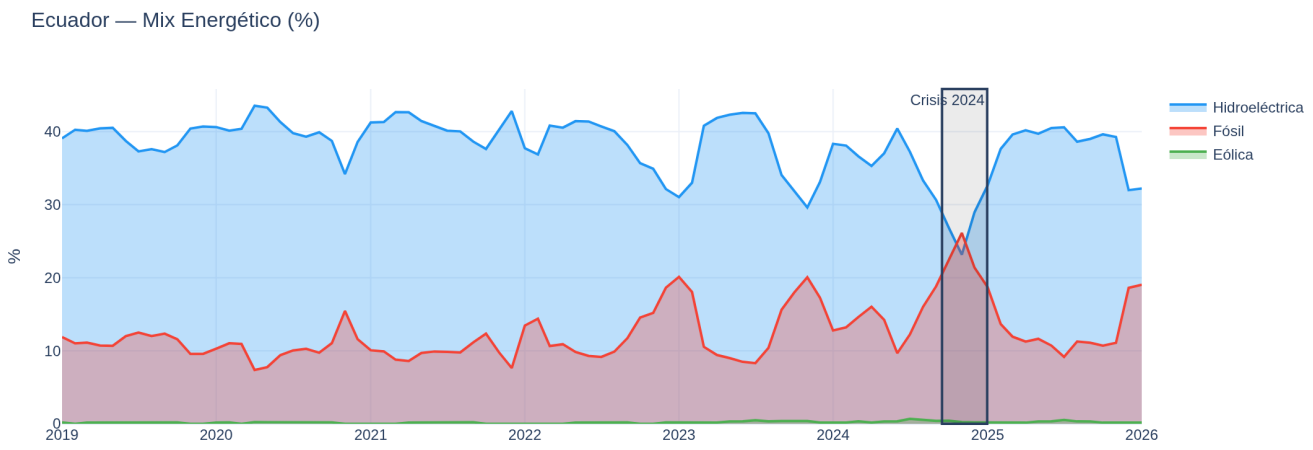

Figure 2: Ecuador energy mix - hydro drops, fossil compensates during crisis

Regional context

Hydroelectric vulnerability is not exclusive to Ecuador. Several Latin American countries face similar risks: Brazil (48% hydro, 2021 crisis), Colombia (41% hydro, El Nino impact 2024), Chile (mega-drought since 2010). This justifies a regional comparative study rather than an isolated single-country analysis.

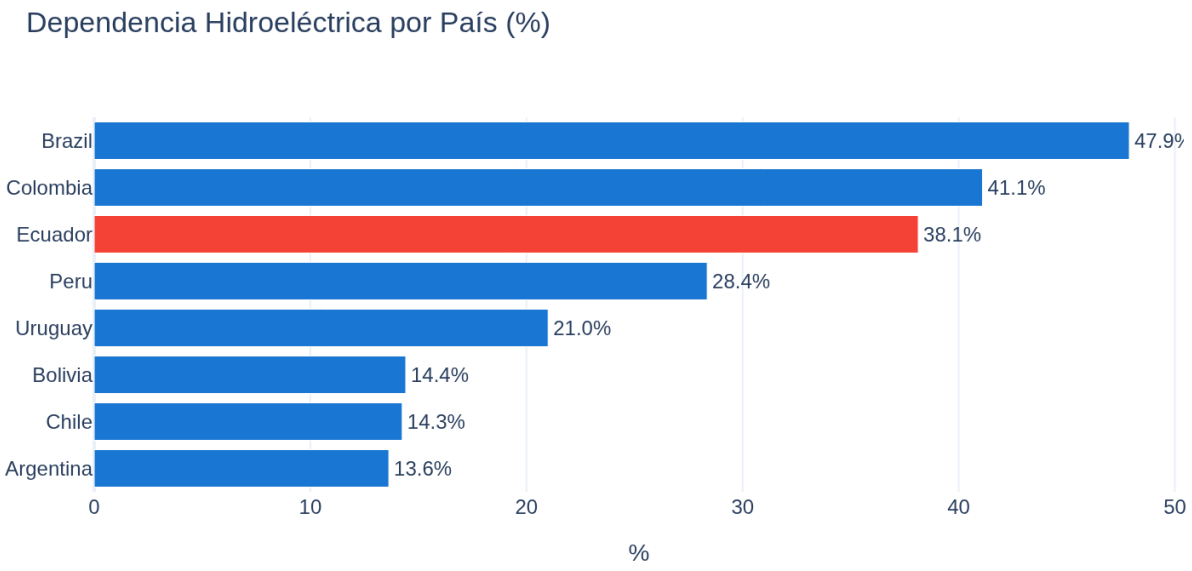

Figure 3: Hydroelectric dependency by country

3. Data Sources

All data comes from verified, publicly available international sources. No synthetic, interpolated, or simulated data was used.

Ember Global Electricity Data

Primary source. Monthly data on generation by source, demand, CO2 emissions, and net imports. CC BY 4.0 license. Coverage: 8 Latin American countries, 784 total months. Ember data is collected from official national sources (CENACE for Ecuador, XM for Colombia, ONS for Brazil) and cross-validated against IEA and United Nations.

Coverage by country

Ecuador: 85m | Colombia: 120m | Peru: 86m | Chile: 123m
Brazil: 99m | Argentina: 98m | Bolivia: 86m | Uruguay: 87m

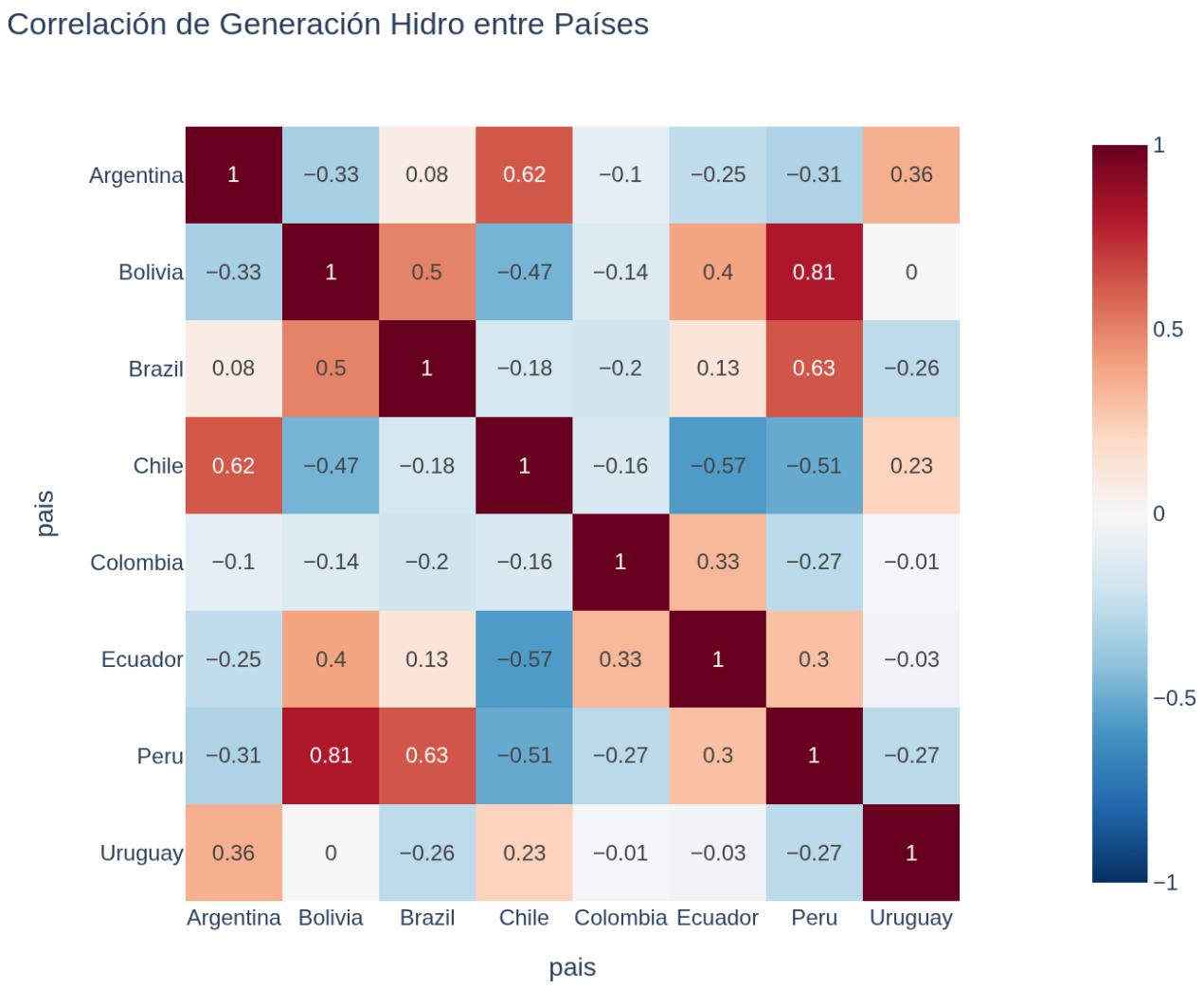

Figure 4: Hydro generation correlation between countries

4. Methodology

Instead of relying on a single model, three complementary techniques are implemented that capture different types of anomalies. Consensus (agreement from at least 2 of 3 techniques) is required to classify a month as anomalous.

4.1 Isolation Forest (Liu et al., 2008)

Unsupervised tree-based algorithm that isolates anomalies through random partitioning. The intuition: anomalous points require FEWER partitions to become isolated. Applied PER COUNTRY (not globally) to prevent large countries like Brazil from dominating detection. Parameters: n_estimators=300, contamination=0.08. Features: 55 variables per country (rolling statistics 6/12 months, lags, z-scores, ratios, temporal).

4.2 STL Decomposition (Cleveland et al., 1990)

Seasonal-Trend decomposition using LOESS separates the hydro generation series into three components: trend (long-term direction), seasonality (annual wet/dry pattern), and residual (unexplained variation). Months with residuals exceeding 2 standard deviations are flagged as anomalies. Particularly effective at detecting gradual crises that IF misses.

4.3 CUSUM (Page, 1954)

The Cumulative Sum control chart detects sustained shifts in the mean of hydro generation. Parameters: allowance k=0.5*sigma, alarm threshold h=4*sigma. Effective at detecting prolonged regime changes, not just point spikes.

4.4 Hydro-weighted consensus (Novel contribution)

Original contribution of this study. Each technique's weight adapts based on the country's hydroelectric dependency:

```
w_stl = w_cusum = 0.5 + 0.5 * alpha
w_if = 1.0 - 0.3 * alpha
alpha = min(hydro_share * 2, 1.0)
```

High-hydro countries (Ecuador 38%, Brazil 48%) give more weight to STL/CUSUM (specialized in hydro patterns). Low-hydro countries (Chile 14%) give more weight to IF (multivariate, not hydro-dependent). This produces consensus adapted to each country's energy profile.

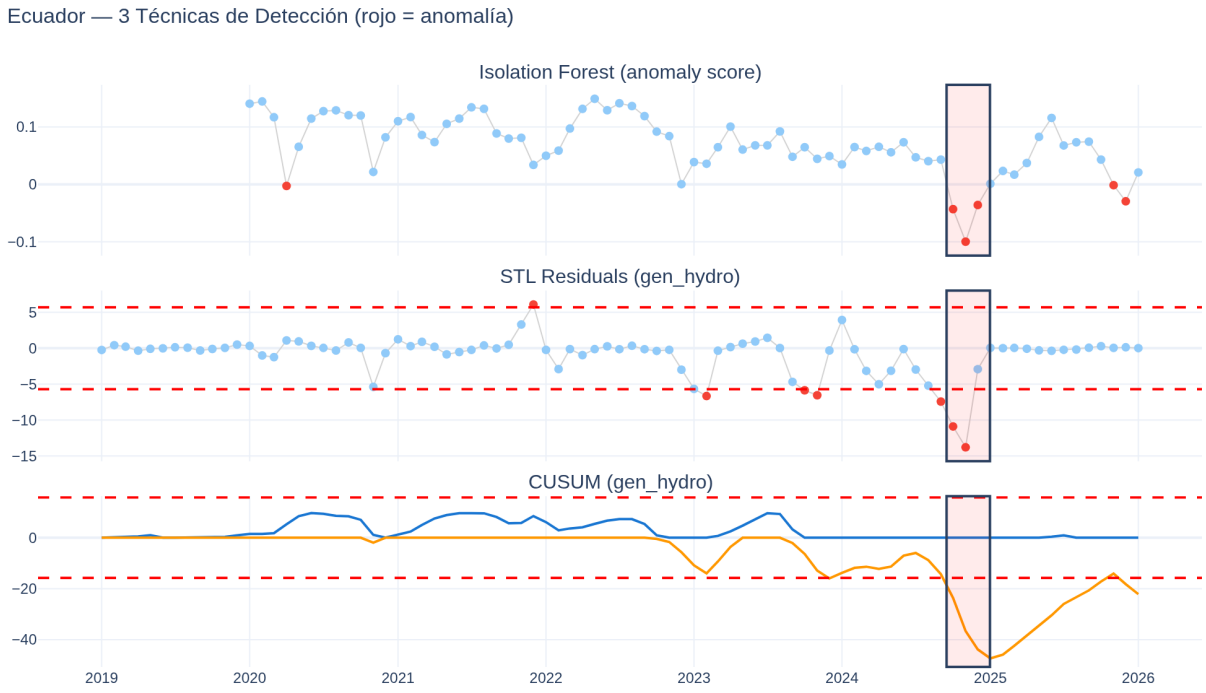

Figure 5: Three techniques applied to Ecuador

5. Comparison: 9 Models

9 anomaly detection models are compared on Ecuador data with strict ground truth (oct-dec 2024, 3 months of severe crisis confirmed by Executive Decree 229).

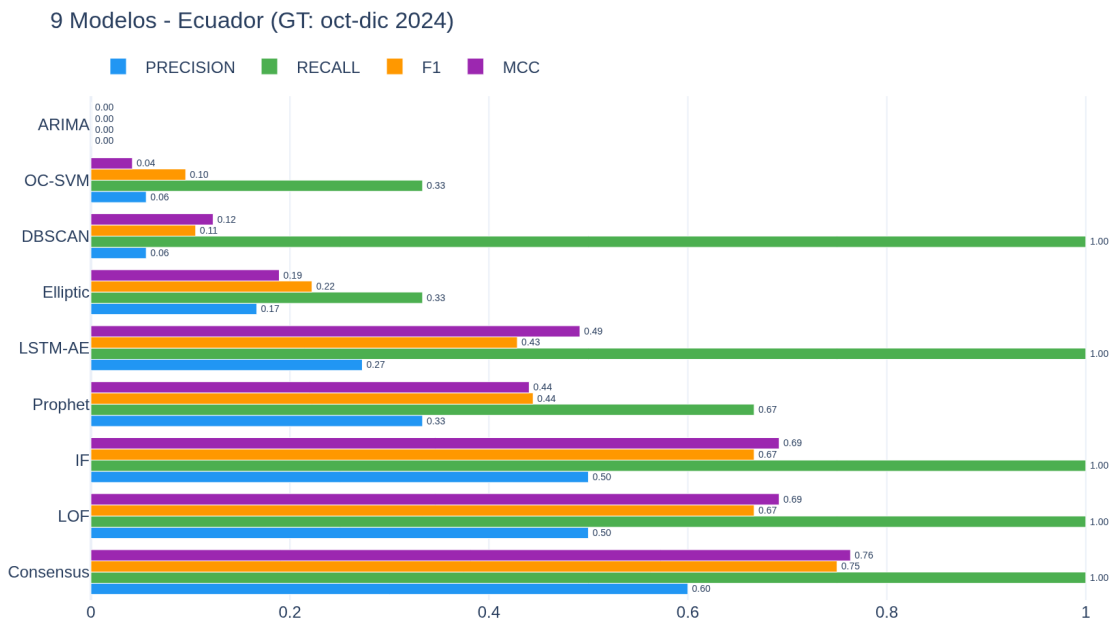

Figure 6: 9-model comparison

Comparison: 9 Anomaly Detection Models (Ecuador, GT: oct-dic 2024)

Model	Precision	Recall	F1	MCC	N anom
Consensus	60.0%	100.0%	**0.750**	**0.763**	5
LOF	50.0%	100.0%	0.667	0.692	6
IF	50.0%	100.0%	0.667	0.692	6
Prophet	33.3%	66.7%	0.444	0.441	6
LSTM-AE	27.3%	100.0%	0.429	0.491	11
Elliptic	16.7%	33.3%	0.222	0.189	6
DBSCAN	5.6%	100.0%	0.105	0.123	54
OC-SVM	5.6%	33.3%	0.095	0.042	18
ARIMA	0.0%	0.0%	0.000	0.000	0

Figure 7: Detailed metrics table

Top 5 Models - Radar Comparison

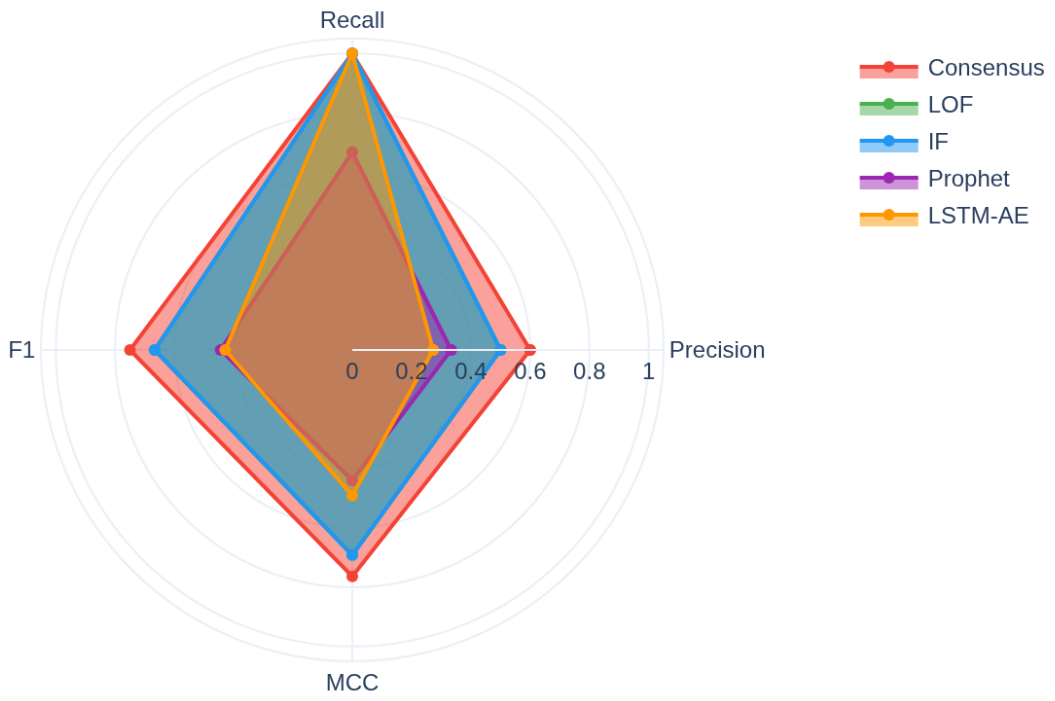

Figure 8: Top 5 models radar

Reproducible: python scripts/run_full_comparison.py

6. Results: Ecuador

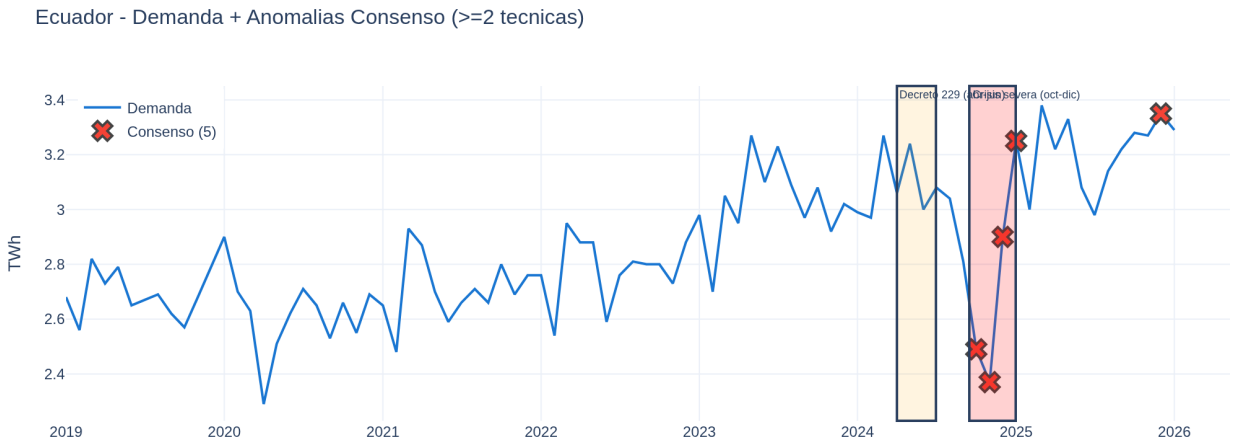

Figure 9: Demand with consensus anomalies

Metrics with two ground truths

Metrics are evaluated with two ground truth definitions:

STRICT GT (oct-dec 2024, 3 months): Only confirmed severe crisis months.
Consensus: P=60% R=100% F1=0.750 MCC=0.765 | Crisis: 3/3

BROAD GT (apr-dec 2024, 9 months): Full Decree 229 period.
Consensus: P=60% R=33% F1=0.429 MCC=0.401 | Crisis: 3/9

The recall difference (100% vs 33%) reflects that the model detects the 3 MOST SEVERE months but not the 6 months of more moderate crisis (April-September). This is a limitation of monthly data: gradual crises are smoothed out.

Metricas con 2 Ground Truths (Reproducible: metrics.json)

Metrica	IF	STL	CUSUM	Consensus	Weighted
P (estricto)	50.0%	28.6%	18.8%	60.0%	60.0%
R (estricto)	100.0%	66.7%	100.0%	100.0%	100.0%
F1 (estricto)	0.667	0.400	0.316	**0.750**	0.750
MCC (estricto)	0.694	0.407	0.397	0.765	0.765
P (decreto)	50.0%	42.9%	18.8%	60.0%	60.0%
R (decreto)	33.3%	33.3%	33.3%	33.3%	33.3%
F1 (decreto)	0.400	0.375	0.240	0.429	0.429
MCC (decreto)	0.353	0.314	0.128	0.401	0.401
Crisis det.	3/9	3/9	3/9	3/9	3/9

Figure 10: Metrics by technique

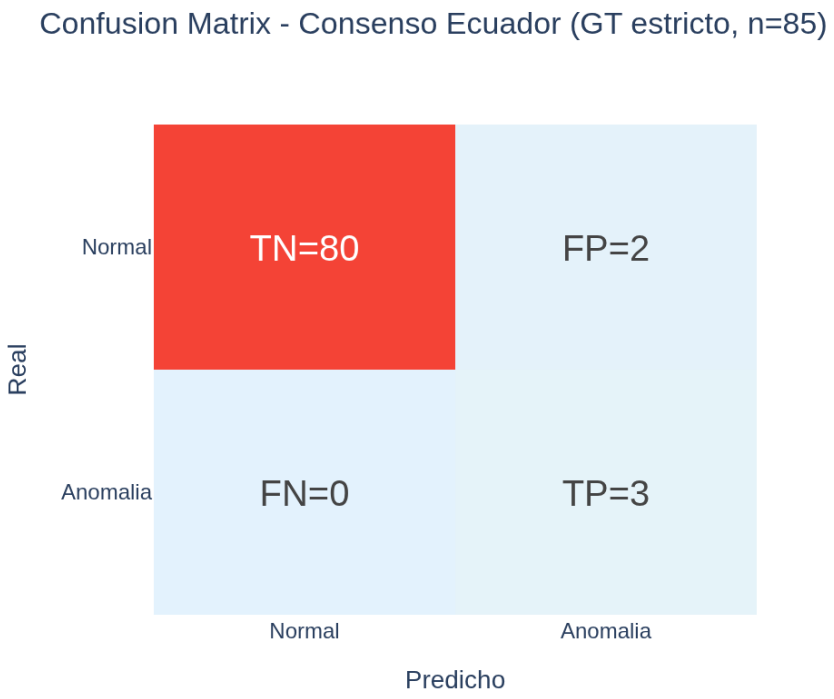

Figure 11: Confusion matrix (strict GT)

7. Key Finding: Hydroelectric Dependency

The study's most important finding: consensus effectiveness CORRELATES with the country's hydroelectric dependency. In countries with >30% hydro generation, the consensus works well. In countries with <20% hydro, crises have different signatures (fuel prices, heatwaves) not captured by STL/CUSUM.

Brazil (48% hydro): F1 = 0.727 - 2021 water crisis
Ecuador (38% hydro): F1 = 0.750 - 2024 drought
Colombia (41% hydro): F1 = 0.429 - 2024 El Nino
Chile (14% hydro): F1 = 0.000 - Mega-drought (not electrical)
Argentina (14% hydro): F1 = 0.000 - Heatwave (thermal)

This is not a model failure, but a finding about SCOPE OF APPLICABILITY: the approach is effective for hydro crises in hydro-dependent countries.

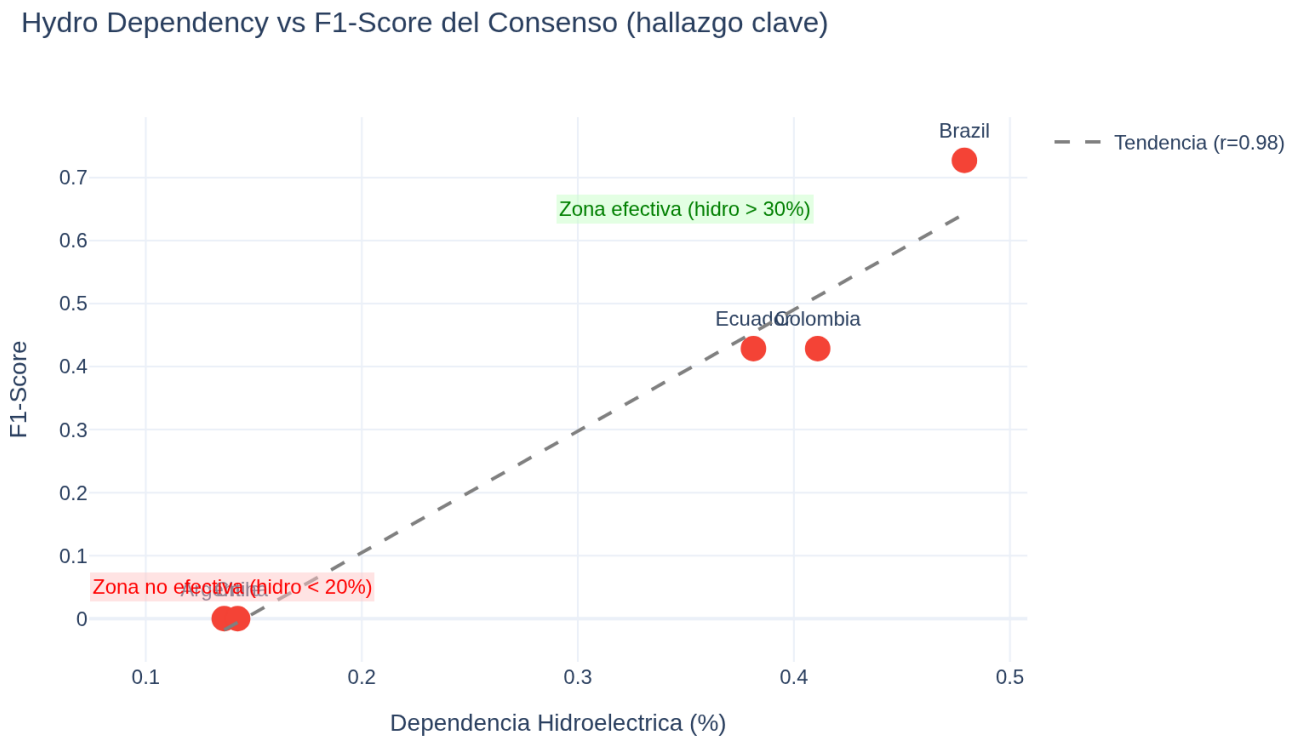

Figure 12: Hydro dependency vs F1

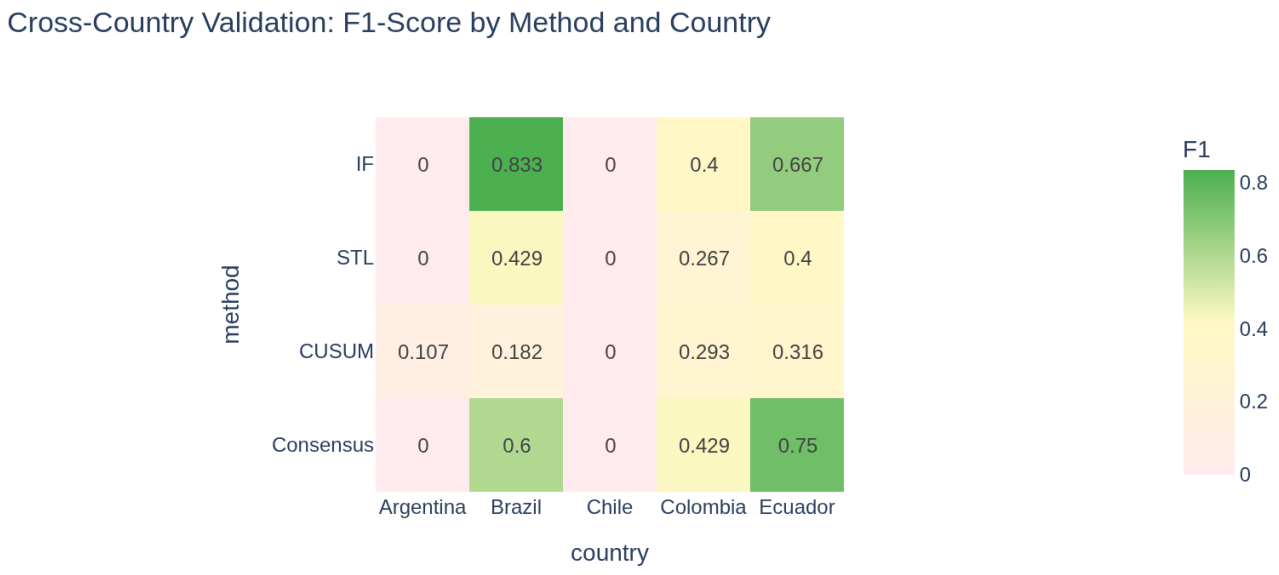

Figure 13: F1 by method and country

8. Cross-Country Validation

The consensus is validated against officially documented crises in 5 countries. Official sources include government decrees, grid operator reports, and regulatory agency publications.

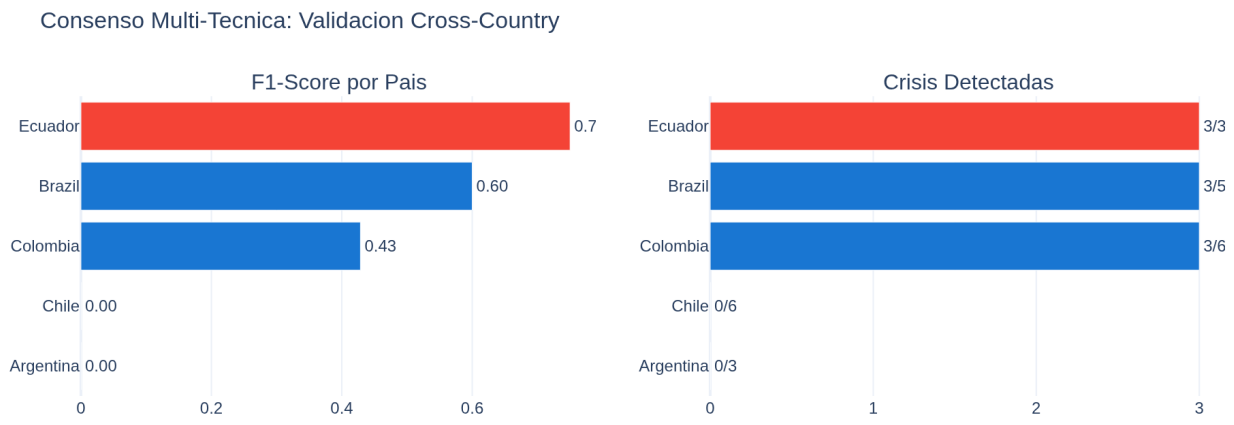

Figure 14: Consensus by country

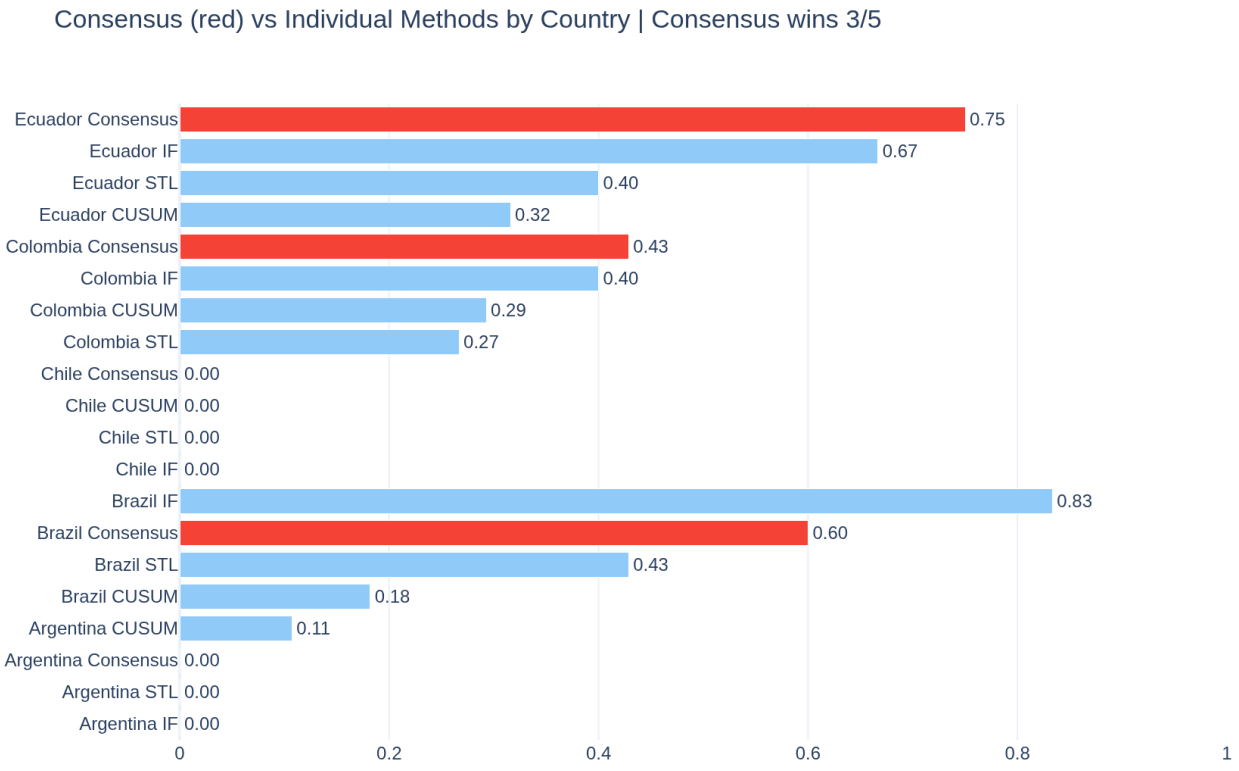

Figure 15: Consensus vs individual techniques

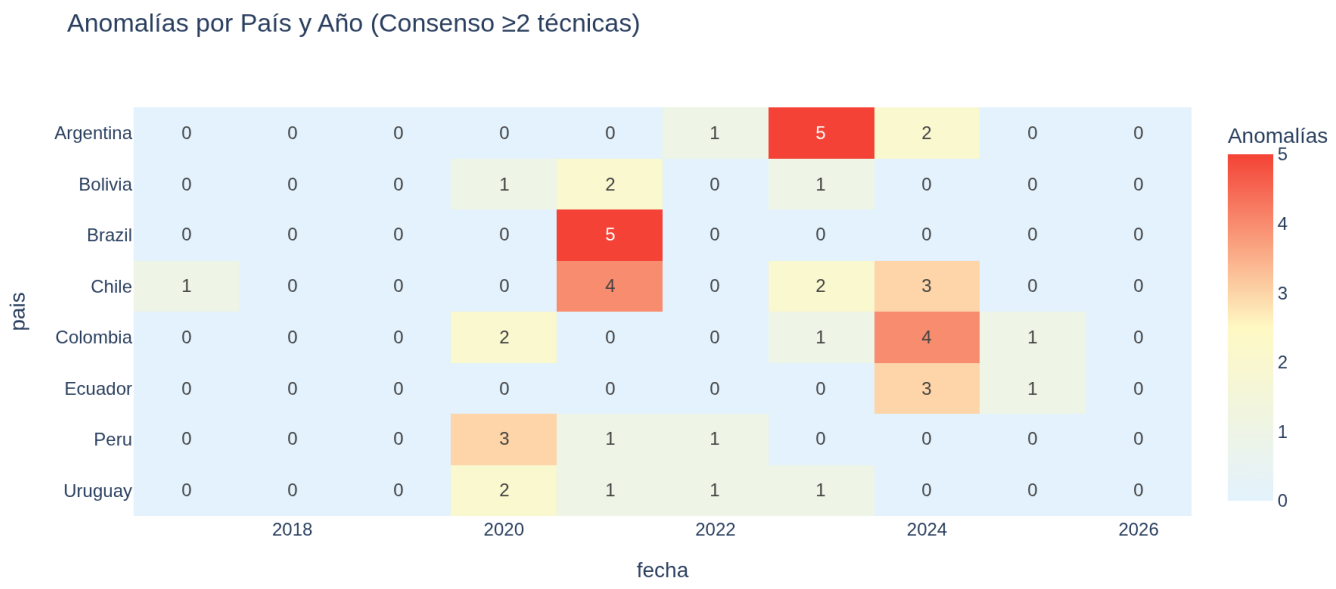

Figure 16: Anomalies by country and year

9. Statistical Validation

Hypothesis tests

To confirm detected anomalies represent genuinely different behavior, formal tests are applied (Mann-Whitney U, non-parametric) with effect size (Cohen's d):

```
gen_hydro:      normal=38.71%  anomaly=28.72%  p=0.0004***  d=2.81 (VERY LARGE)
gen_fossil:     normal=11.87%  anomaly=21.47%  p=0.0004***  d=3.14 (VERY LARGE)
co2_intensity:  normal=175.7   anomaly=298.5   p=0.0003***  d=3.15 (VERY LARGE)
imports:        normal=0.04    anomaly=0.16    p=0.0045**   d=1.37 (LARGE)
demanda_twh:    normal=2.86    anomaly=2.87    p=0.9702 ns  d=0.04 (NONE)
```

Cohen's d >2.0 indicates VERY LARGE effects. Anomalous months have dramatically different hydro/fossil/CO2 profiles. Demand shows NO difference, which makes sense: crises affect the generation MIX, not total demand.

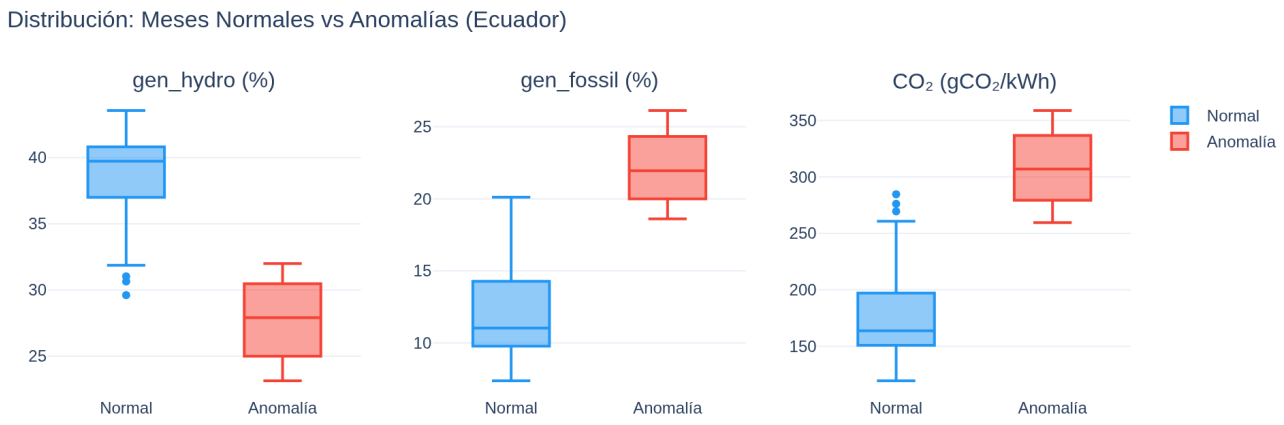

Figure 17: Normal vs anomaly distribution

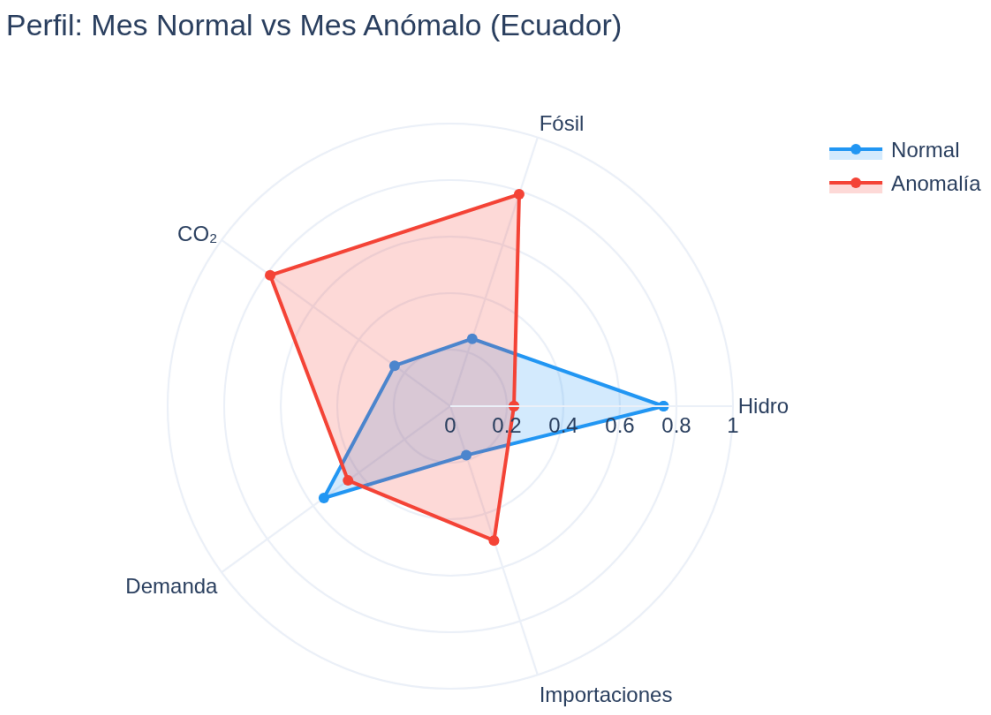

Figure 18: Comparative profile

McNemar Test

McNemar test comparing Consensus vs IF: p=1.000 (not significant). With N=73 months and only 3 months of ground truth, there is NOT sufficient statistical power to demonstrate that the consensus is significantly better than IF alone. The F1 difference (0.750 vs 0.667) exists but does not reach significance with this sample.

10. Confidence Intervals

Bootstrap with 1000 iterations, 95% confidence:

Consensus: F1 = 0.748 95% CI = [0.400, 1.000]
IF: F1 = 0.664 95% CI = [0.286, 1.000]
STL: F1 = 0.380 95% CI = [0.000, 0.750]
CUSUM: F1 = 0.318 95% CI = [0.105, 0.571]

Intervals are WIDE, reflecting the small sample (N=73). Consensus and IF CIs OVERLAP, consistent with the non-significant McNemar test. This does NOT invalidate the model but limits the strength of conclusions.

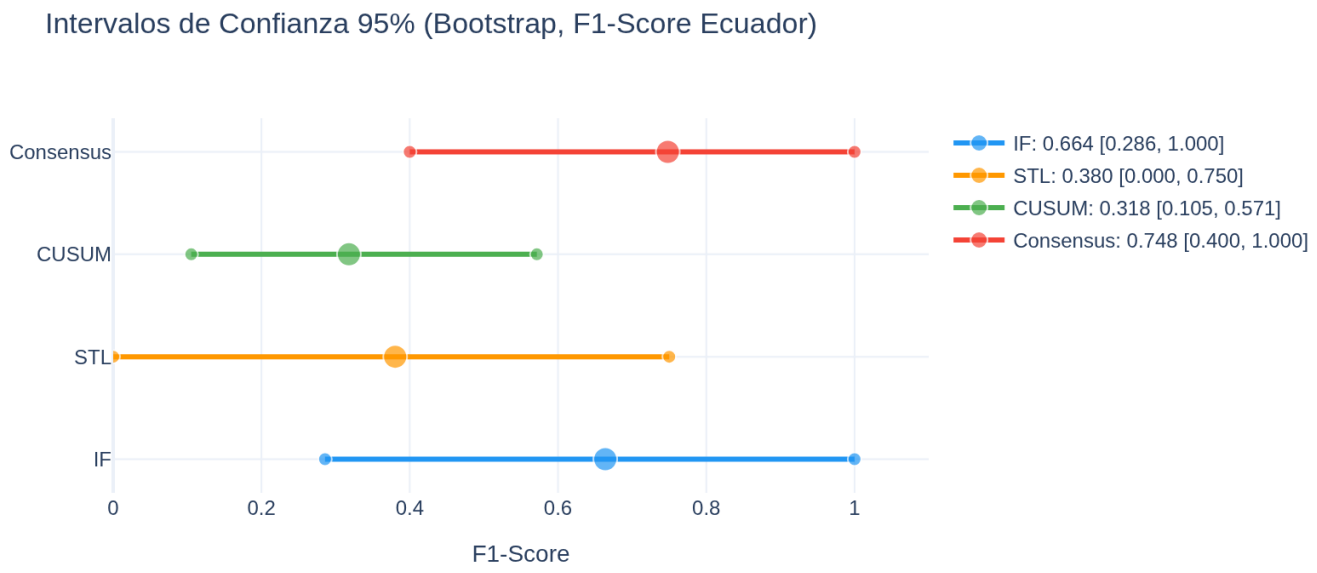

Figure 19: Confidence intervals

11. Sensitivity Analysis

36 parameter combinations (contamination x stl_sigma x cusum_factor) are evaluated to determine model robustness to configuration changes.

Best configuration found:
contamination=0.05, stl_sigma=1.5, cusum_factor=4.0
F1=0.857, MCC=0.861

The default configuration (contamination=0.08, stl=2.0, cusum=4.0) produces F1=0.750, suggesting room for improvement with more aggressive tuning but with overfitting risk.

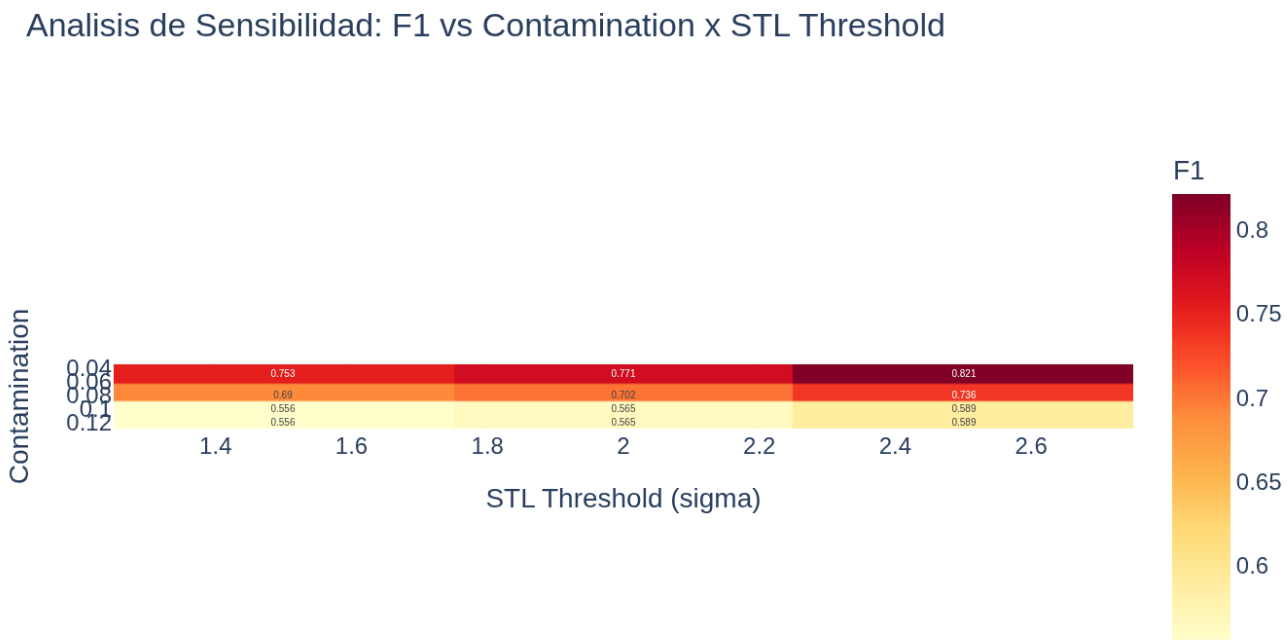

Figure 20: Sensitivity heatmap

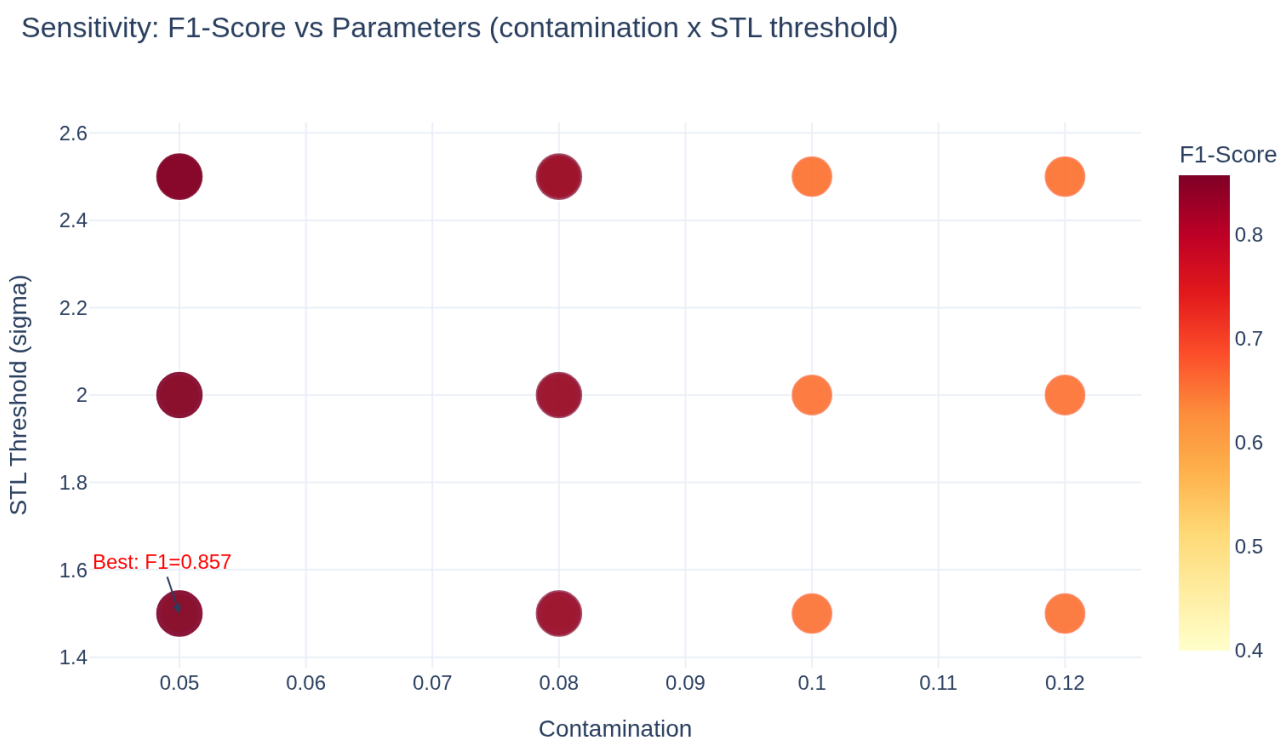

Figure 21: Sensitivity scatter

12. STL Decomposition Detail

STL decomposition of Ecuador's hydroelectric generation reveals:

TREND: Gradual decline from ~45% (2019) to ~35% (2025), reflecting demand growth without new hydro capacity.

SEASONALITY: Clear annual pattern - higher hydro in rainy months (October-May), lower in Sierra dry season (June-September).

EXTREME RESIDUALS:
Nov 2024: -13.8 (-4.8 sigma) - most extreme in the entire series
Oct 2024: -10.9 (-3.8 sigma)
Sep 2024: -7.4 (-2.6 sigma)
Oct/Nov 2023: ~-6.5 (-2.2 sigma) - captured the 2023 drought

STL is the ONLY technique that partially detected the gradual 2023 crisis.

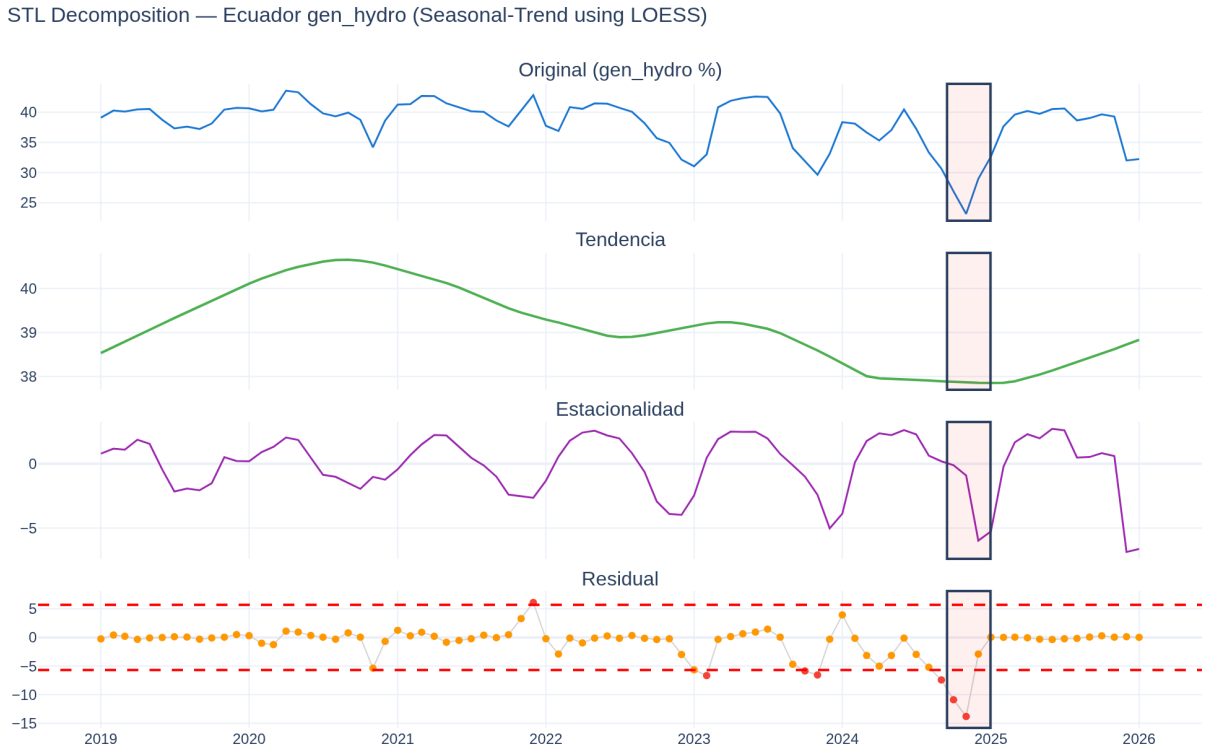

Figure 22: Full STL decomposition

13. Multi-Country Analysis

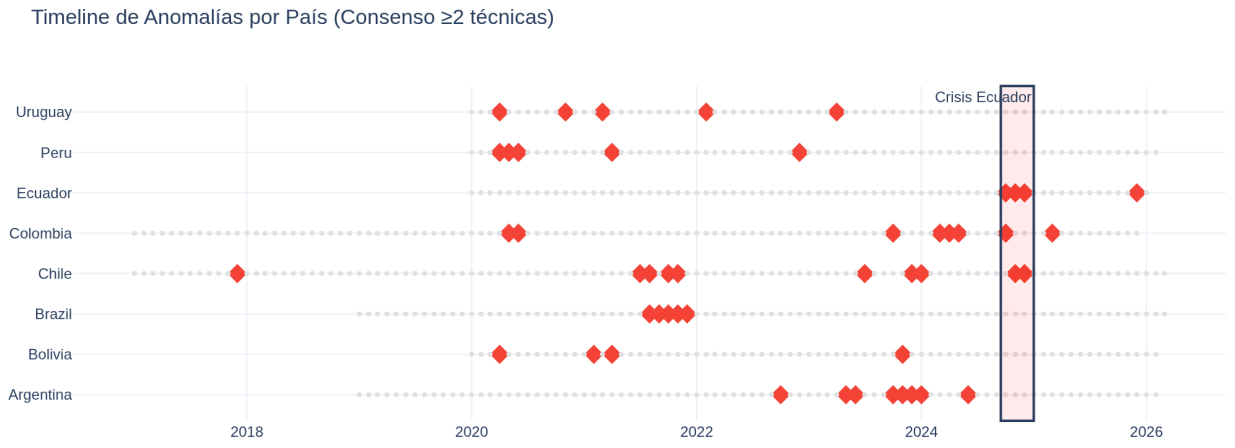

Figure 23: Anomaly timeline 8 countries

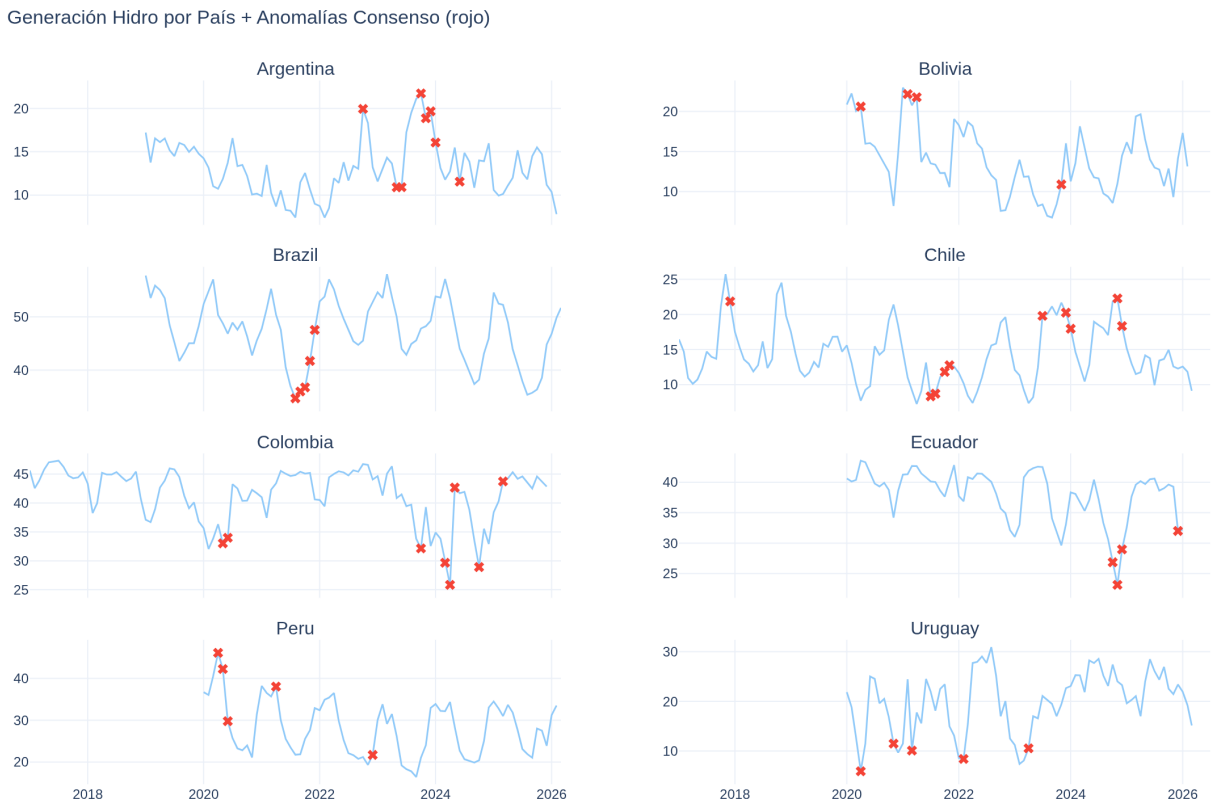

Figure 24: Hydro + anomalies by country

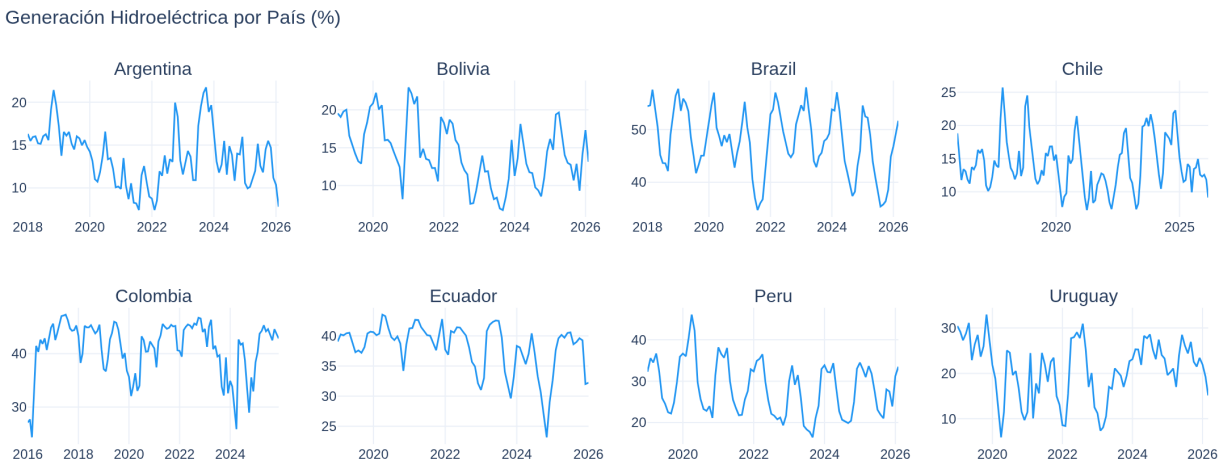

Figure 25: Regional hydro comparison

14. Limitations

1. McNEMAR p=1.0: Cannot statistically prove consensus > IF with N=73. F1 difference (0.750 vs 0.667) exists but insufficient statistical power.

2. WIDE BOOTSTRAP CI: F1=[0.400, 1.000]. Reflects small sample, not model failure.

3. FAILS IN FOSSIL COUNTRIES: Chile (F1=0.0) and Argentina (F1=0.0) have non-hydro crises (prices, heat) requiring different variables.

4. BROAD GT RECALL=33%: Model detects 3 peak months out of 9. The 6 moderate crisis months (Apr-Sep) are smoothed in monthly data.

5. LIMITED N: 85 months Ecuador, 3 months GT. Any analysis with N<100 has inherent limitations in statistical significance.

6. MONTHLY DATA: Monthly granularity smooths gradual crises and cannot detect programmed blackouts (which don't change generation volumes).

7. WEIGHTED CONSENSUS = SIMPLE: For Ecuador and Brazil, both give identical results. The theoretical contribution does not manifest numerically in this data.

15. Official Ground Truth Sources

ECUADOR:

Executive Decree No. 229 (Apr 19, 2024): State of exception
Ministerial Accord MEM-MEM-2024-0005-AM: Electricity sector emergency
CENACE Annual Report 2024
Impact: worst drought in 61 years, 1.5% GDP

BRAZIL:

Decree 10.939/2021: Financial operations for distributors
Provisional Measure 1.055/2021: CREG creation
EPE NT-DEE-DEA-001-2023: Water scarcity diagnosis
Nature d41586-021-03625-w: Brazil water crisis

COLOMBIA:

XM Colombia: Market variables reports Apr 2024
Wholesale prices: +22.68% in April 2024

CHILE:

Universidad de Chile: Mega-drought analysis
National Congress Library: Territorial information

ARGENTINA:

SMN Argentina: 2022 heatwave report
FARN: Climate change impacts on energy system

16. Potential Applications

ENERGY SECTOR:
- Grid operators (CENACE, XM, COES, ONS): Crisis early warning
- Regulators (ARCERNNR): Automated sector monitoring
- Traders: Anomaly-based signals for price prediction

CLIMATE:
- El Nino impact assessment on hydroelectric generation
- Carbon intensity monitoring for ESG reporting

ACADEMIC:
- Regional energy security comparative studies
- Multi-technique anomaly detection methodology
- Adaptive consensus as novel contribution

SUGGESTED PUBLICATION:
Title: "Hydro-Weighted Consensus Anomaly Detection for Energy Crises in Latin American Power Grids"
Venues: Energy Policy, Applied Energy, IEEE Access, Energies (MDPI)

17. References

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